

Arduino-Based SMARTPHONE CONTROLLED ROBOT CAR

DR P.S. MANOHARAN AND S. NAVEEN

Nowadays smartphones can be used to control a host of electrical and electronic devices including motors, music

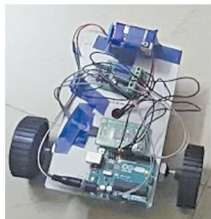


Fig. 1: Authors' prototype

systems and lights. Here we present an Arduino based robot car, which can be controlled using an Android smartphone having ArduinoRC application installed in it. This bot receives commands from your smartphone with the help of a Bluetooth module. The authors' prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of Arduino-based smartphone-controlled car is shown in Fig. 2. Pin mapping of the Arduino Uno is shown in Fig. 3.

The ATmega328 chip on the Arduino Uno board comes with a bootloader, which allows you to burn the program to the chip without an ex-

ternal hardware programmer. After the chip is programmed, it can be removed from the board. The board can be powered from USB connection or external power supply such as batteries or AC/DC adaptors. After connecting the power supply to Vin pin, you get regulated 5V and 3.3V DC on the respective pins of the Arduino board. 5V is used to power the HC-05 Bluetooth module.

Serial pins 0 (RX) and 1 (TX) of Arduino are used to communicate with HC-05 module. Pins 4 through 7 are connected to H-bridge motor driver L293D IC. Out of these four pins, pins 5 and 6 provide 8-bit PWM output using analogWrite() function defined in the asc_car.ino Arduino sketch. Pins 2 and 3 of Arduino (not used in this project) can be used for additional functions like controlling lights, horn, etc.

Construction and testing

A 9V battery supply is used to power the Arduino Uno with its positive terminal connected to Vin pin and negative terminal connected to GND pin. 5V and GND (ground) pins from the Arduino are connected to HC-05 Bluetooth module. RX (pin 0) and TX (pin 1) pins of the Arduino are connected to TX and RX pins of the HC-05 module, respectively.

Pins 4 through 7 of the Arduino provide controlling signals for the motors. The Arduino outputs a very small voltage from these pins. So, an additional power supply is required to produce enough torque for the motors, for which H-Bridge L293D module is used. You can provide additional 12V power supply to pin 8 (VCC2) of L293D to

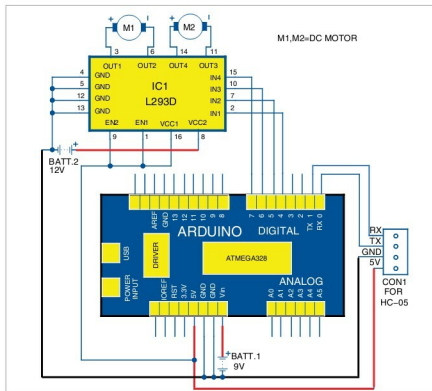


Fig. 2: Circuit diagram of Arduino-based smartphone-controlled car